

IGZO TFT-Based All-Enhancement Operational Amplifier Bent to a Radius of 5 mm

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Abstract—An all-enhancement operational amplifier operating at 5 V and comprising 16 n-type amorphous indium–gallium–zinc–oxide thin-film transistors (TFTs) is fabricated on a 50 μm thick flexible polyimide substrate. The operational amplifier has an open loop voltage gain of 18.7 dB and a unity-gain frequency of 472 kHz while the common-mode rejection ratio (CMRR) is larger than 40 dB. The mechanical flexibility of the amplifier is demonstrated by bending the circuit to a radius of 5 mm, which corresponds to a tensile strain of 0.5% parallel to the TFT channels. The bent amplifier shows the same output behavior as when flat. The power consumption of the operational amplifier is 900 μW , regardless whether the circuit is flat or bent.

Index Terms—Analog circuits, indium–gallium–zinc–oxide (IGZO), operational amplifiers, thin-film circuits, thin-film transistors (TFTs).

I. INTRODUCTION

BENDABLE transistors and circuits are the key components for flexible electronic applications, such as flexible displays, sensor skins, or electronic textiles [1]. To reduce integration efforts and production costs of flexible electronic systems, the system-on-panel (SoP) approach aims at combining sensors and actuators, such as individual pixels of a display, on flexible substrates with the required peripheral control electronics, as demonstrated in [2]. To realize flexible SoPs, analog circuits and, in particular, operational amplifiers are required to amplify sensor signals and to convert analog signals to digital signals or vice versa.

There are several promising semiconductor material classes for flexible electronics: amorphous silicon [3], organic semiconductors [4], submicrometer-thin layers of single-crystalline silicon [5], or oxide semiconductors [6]. For all these classes, analog circuits are demonstrated, but not all of the circuits are flexible. In [7], an amorphous silicon operational amplifier, as a part of a 4-b digital-to-analog converter on a glass substrate is reported. At an operation voltage of 25 V, the amorphous silicon operation amplifier exhibits a gain of 42.5 dB and a unity-gain frequency of 30 kHz. An organic semiconductor-based pseudo-CMOS amplifier fabricated on a glass substrate is demonstrated in [4]. For the amplifier, a voltage gain of

41.8 dB and a unity-gain frequency of 5 kHz are specified. Ahn *et al.* [8] reported an operational amplifier based on printed ribbons of single-crystalline silicon on a bendable polyimide substrate. The operational amplifier has a voltage gain of 4 dB, a unity-gain frequency of ≈ 100 kHz and can be bent to a radius of 6 mm, which corresponds to a strain of 0.23%. In [9], the oxide–semiconductor indium–gallium–zinc–oxide (IGZO) is used to fabricate flexible common-source amplifiers on a polyimide substrate. The common-source amplifiers exhibit a gain of 6.8 dB and -3 dB frequency of 1.2 MHz.

In this letter, we present a mechanically flexible all-enhancement operational amplifier directly fabricated on a 50 μm thick polyimide substrate. As semiconductor, IGZO was used because of its electron mobility >10 cm^2/Vs when deposited at room temperature. This enables thin-film transistor (TFT) fabrication with a maximum process temperature of 150 $^\circ\text{C}$ suitable for flexible plastic substrates [10]. The saturation mobility of the IGZO TFTs for the operational amplifier exceeds 18 cm^2/Vs and the threshold voltage is 0.9 V. For the operational amplifier, an open-loop voltage gain of 18.7 dB and a unity-gain frequency of 472 kHz were measured at a supply voltage of 5 V. Furthermore, bending the circuit to a radius of 5 mm has virtually no effect on the open-loop voltage gain or the unity-gain frequency.

II. FABRICATION AND CIRCUIT DESIGN

IGZO TFTs and operational amplifiers based on IGZO TFTs were fabricated on a free-standing 50 μm thick polyimide foil (Kapton). Fig. 1(a) shows a cross section of the fabricated IGZO TFT. A 50 nm thick Plasma-enhanced chemical vapor deposition deposited silicon nitride (SiN_x) layer serves as adhesion layer. The gate contact consists of an e-beam evaporated 35 nm thick Cr layer, structured using photolithography. The 25 nm thick aluminum oxide (Al_2O_3) gate isolator was fabricated using atomic layer deposition with the highest temperature within the process of 150 $^\circ\text{C}$. Next, 15 nm of IGZO was RF-magnetron sputtered at room temperature and structured using photolithography. To contact the gate electrodes and create the vertical interconnects for circuits, openings were etched into the Al_2O_3 layer. For drain and source contacts, 10 nm titanium and 60 nm gold were e-beam evaporated and structured using liftoff. Finally, the whole devices were passivated with a 25 nm thick Al_2O_3 layer.

Fig. 1(c) shows the schematic diagram of the fabricated operational amplifier. The circuit consists of a differential

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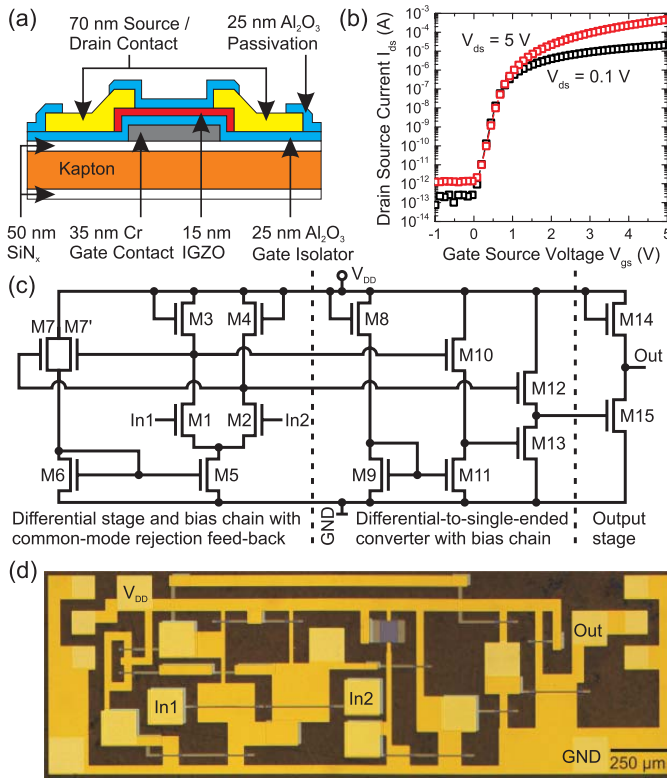


Fig. 1. (a) Cross section of fabricated IGZO TFT. (b) Transfer characteristic of IGZO TFT with W/L of 50/6 μm . (c) Schematic diagram of the IGZO TFT-based all-enhancement operational amplifier. M1–M5: the differential stage. M6, M7, and M7': the bias chain for M5 including a common-mode suppression. M8–M13: a differential-to-single-ended converter with M8 and M9 as the bias stage. M14 and M15: the output stage. (d) Optical microscope image of the fabricated amplifier.

stage with a common-mode rejection feedback, a differential-to-single-ended converter and an output stage, as described in [11]. The design was verified using the level 61 model in HSpice (AIM spice level 15 model) with the simulation parameters for the IGZO TFTs obtained according to the procedure proposed in [12].

All the TFTs have a channel length of 6 μm with the exception of M8, which has a channel length of 90 μm . A channel length of 6 μm was chosen because it can be reliably fabricated with the process described in the previous section. M1 and M2, which are the input TFTs for the differential stage, have a channel width (W) of 245 μm and the corresponding load transistors (M3 and M4) in the differential stage are designed with $W = 15 \mu\text{m}$. The W/L ratios for M1–M4 were set to allow a differential voltage amplification of ≈ 2.5 with a supply voltage of 5 V, a dc input voltage of 2.5 V, and a dc output voltage of the differential stage of ≈ 3.5 V. The channel width of the TFTs M5 and M6 of the current source is 67 μm , whereas TFTs M7 and M7', which increase the common-mode rejection, have a W of 31 μm . TFTs M10 ($W = 30 \mu\text{m}$), M11 ($W = 97 \mu\text{m}$), M12 ($W = 30 \mu\text{m}$), and M13 ($W = 32 \mu\text{m}$) convert the dual-ended output signals of the differential stage into a single-ended signal. TFTs M8 ($W = 15 \mu\text{m}$) and M9 ($W = 97 \mu\text{m}$) form the bias chain for the differential-to-single-ended converter. Finally, TFTs M14 ($W = 30 \mu\text{m}$) and M15 ($W = 177 \mu\text{m}$)

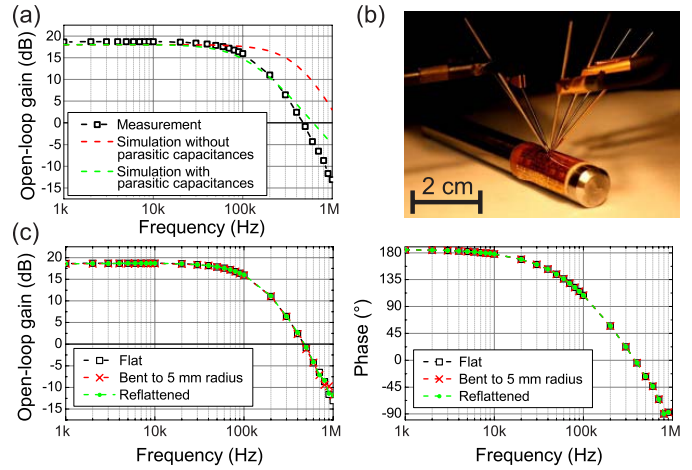


Fig. 2. (a) Frequency response of the operational amplifier compared with the simulation of the circuit. The red dashed line shows the simulation result without parasitic capacitances, which occur because of crossing lines within the fabricated circuit, whereas the simulation result with the green dashed line includes the parasitic capacitances. (b) Photograph of the bent and contacted circuit. (c) Frequency responses of the amplifier before bending, bent to a radius of 5 mm (0.5% strain), and reflattened after bending.

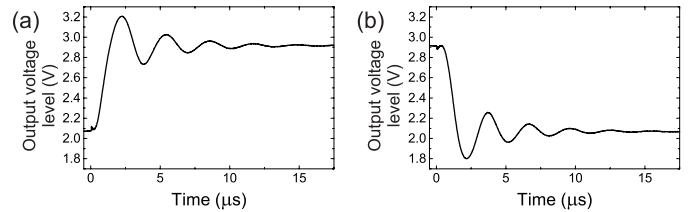


Fig. 3. Rising and falling edges of the operational amplifier configured as unity-gain buffer. The input voltage step of 1 V is applied at 0 μs .

form the output stage of the operational amplifier. Fig. 1(d) shows a photograph of the fabricated operational amplifier, which covers an area of 2.6 mm $\times$ 0.8 mm.

III. RESULTS AND DISCUSSION

Single-IGZO TFTs were characterized using a semiconductor parameter analyzer (Agilent B1500A). Operational amplifiers were tested by applying a differential sinusoidal signal with a peak-to-peak amplitude of 100 mV, generated with an Agilent 33522A dual-channel signal generator to the inputs, while a supply voltage of 5 V was applied. The output was contacted with a load of 2 pF and 1 M Ω and monitored with an oscilloscope (Agilent MSO-X-3034A).

For the fabricated TFTs, an electron mobility of 18.3 cm²/V s, a transconductance of 6 $\mu\text{S } \mu\text{m}^{-1}$, a threshold voltage of 0.9 V, an ON/OFF ratio $> 10^8$ and a contact resistance of 66 k $\Omega \mu\text{m}$ was extracted using standard transistor equations [13]. A transit frequency of 11.5 MHz was measured using a network analyzer (HP 8753E). A transfer characteristic for an IGZO TFT with a W/L of 50/6 μm is shown in Fig. 1(b). Applying tensile strain of 0.5% parallel to the channel length of the TFT by attaching the TFT to a cylindrical rod with a radius of 5 mm increased the electron mobility by $\approx 2\%$ [14].

Fig. 2(a) shows the frequency response of the all-enhancement operational amplifier based on IGZO TFTs

TABLE I
OPEN-LOOP GAIN AND UNITY-GAIN FREQUENCY OF THE OPERATIONAL
AMPLIFIER WHILE FLAT, BENT TO A RADIUS OF
5 mm AND REFLATTENED

	Open-loop gain (dB)	Unity-gain frequency (kHz)
Flat	18.7	472
Bent ($r = 5$ mm)	18.6	465
Reflattened	18.7	476

(black line). An open-loop voltage gain of 18.7 dB and a unity-gain frequency of 427 kHz were measured at an operation voltage of 5 V. The unity-gain bandwidth of the amplifier is limited by the differential stage. The power consumption of the amplifier was 900 μ W. Up to a frequency of 108 kHz (corresponding to the -3 dB frequency), the common-mode CMMR was >40 dB. At the unity-gain frequency, CMMR was >20 dB. An open-loop output swing of 3 V ($V_{\text{low}} = 0.6$ V and $V_{\text{high}} = 3.6$ V) was measured. Slew rates of 0.88 V μ s $^{-1}$ (up) and 1.1 V μ s $^{-1}$ (down) were measured by applying a 1 V step to the operational amplifier contacted as a unity-gain buffer (closed-loop configuration). The corresponding rising and falling edges are shown in Fig. 3(a) and (b).

In addition, Fig. 2(a) shows the simulation output (dashed lines). It is worth noting that the measured and the simulated gain match. The simulation results show that an operational amplifier with no parasitic capacitances yields a unity-gain frequency above 1 MHz (red dashed line). However, because of crossing lines within the fabricated circuit [Fig. 1(c)], internal signal lines are loaded with additional capacitances. By including these parasitic capacitances (1–3 pF per crossing because of the 25 nm thick Al_2O_3 layer with an $\epsilon_r = 9.5$) in the simulated circuit, simulation and measurement are in good agreement, as shown by the dashed green line (unity-gain frequency of ≈ 600 kHz).

To test the mechanical flexibility of the IGZO TFT-based all-enhancement operational amplifier, the circuit was attached to a cylindrical rod with a diameter of 5 mm (0.5% strain), as shown in Fig. 2(b). Strain was applied parallel to the channel of the TFTs. The frequency responses for the flat circuit, the circuit bent to a radius of 5 mm, and finally, reflattened are shown in Fig. 2(c) and summarized in Table I.

As shown in Table I, applying strain has virtually no influence on the open-loop gain. The small variations in the unity-gain frequency are due to recontacting the circuit, and therefore changing the capacitive load. Despite the fact that strain changes the electron mobility μ and therefore influences linearly the transconductance g_m and drain–source conductance g_d of the TFTs, the strain independence of the open-loop gain was expected. First, amplifications of all the stages of the amplifier are described by ratios of g_{mi}/g_{mj} or g_{mi}/g_{dj} [11], where the strain-dependent mobility gets canceled out. Second, as shown in Fig. 1(d), all the TFT channels are aligned in the same direction and are hence equally influenced by the strain.

IV. CONCLUSION

In this letter, to the best of our knowledge, we have presented the first mechanically flexible all-enhancement

operational amplifier based on IGZO TFTs and directly fabricated on a flexible foil substrate. The open-loop voltage gain of the operational amplifier is 18.7 dB and the unity-gain frequency 472 kHz at a 5 V operation voltage. Bending the operational amplifier to a radius of 5 mm (0.5% strain) did not influence the frequency response of the circuit.

To improve the frequency behavior of the flexible operational amplifier, several strategies are applicable. First, reducing the size of the crossing lines decreases parasitic capacitances within the circuit. Second, using a self-alignment process to fabricate the circuit reduces the overlap capacitances between gate and drain, and gate and source increasing the switching speed of the individual TFTs [15].

These findings are an important step toward the realization of flexible electronic systems, which require sensor signal amplification and analog-to-digital conversion, such as flexible sensors for electronic textiles.

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